

Measurement of the neutrino magnetic moment at the NOvA experiment

Technical note

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Abstract

This is the abstract

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1 Introduction

In this analysis, we aim to detect a potential signal of the effective muon neutrino magnetic moment in the NuMI Off-axis ν_e Appearance (NOvA) Near Detector (ND). This signal would manifest as an excess of neutrino-on-electron (ν -on- e) elastic scattering interactions at low electron recoil energies, proportional to the value of the effective neutrino magnetic moment, over the Standard Model (SM) background. If no significant excess is observed, we will establish an upper limit on the effective muon neutrino magnetic moment.

Detecting the neutrino magnetic moment (μ_ν) would provide definitive evidence of new Beyond Standard Model (BSM) physics, and measuring its value would help identify the appropriate BSM theory. As current and planned experiments can only detect an anomalously large neutrino magnetic moment, observing such a signature would strongly suggest that neutrinos are Majorana particles and would have significant implications for astrophysics and cosmology [1].

The best model-independent experimental results on the neutrino magnetic moment come from experiments searching for dark matter using xenon-based detectors. These highly sensitive detectors detect solar neutrinos, which are part of the background in dark matter searches but can be reanalysed for other purposes. In 2020, the XENON1T experiment observed [2] a low energy excess of solar neutrinos, which could correspond to a signal from an anomalously large effective magnetic moment within $\mu_{\nu_\odot} \in (0.14, 0.29) \times 10^{-10} \mu_B$ at 90 % Confidence Level (C.L.), where $\nu_\odot$ marks solar neutrinos. However, this result was disfavoured by the follow-up XENONnT experiment in 2022 [3], which saw no excess and set the current world-leading limit on neutrino magnetic moment at $\mu_{\nu_\odot} < 0.063 \times 10^{-10} \mu_B$ at 90 % C.L.. Other solar neutrino experiments also reported null results regarding neutrino magnetic moment [4, 5], placing less stringent limits on its value. Given some basic assumptions [5, 6] this limit for solar neutrinos would correspond to a limit on muon neutrino effective magnetic moment of $\mu_{\nu_\mu} < 0.137 \times 10^{-10} \mu_B$. However, the relationship between effective magnetic moments of different neutrino flavours may be non-trivial, especially in the context of possible new BSM physics, and studying muon neutrinos remains an important endeavour [7].

The best results for ν_μ and $\bar{\nu}_\mu$ come from accelerator-based stopped pion neutrino sources [8, 9], which also do not observe any low energy excess and provide an upper limit on the effective muon neutrino magnetic moment of $\mu_{\nu_\mu} < 6.8 \times 10^{-10} \mu_B$ at 90 % C.L. [8]. Stopped pion neutrino sources provide well-understood beams made up of ν_μ , $\bar{\nu}_\mu$ and ν_e with energies up to 52.8 MeV. Slightly looser limits come from pion decay-in-flight accelerator-based measurements (similar to NOvA) [10, 11], which provide a limit of $\mu_{\nu_\mu} < 8.5 \times 10^{-10} \mu_B$ at 90 % C.L..

Thanks to the very intense and highly pure beam of muon neutrinos and antineutrinos, and a detector designed for the reconstruction and identification of events with electrons in the final state, NOvA is well-positioned to provide a highly competitive, and possibly even world-leading, measurement (or limit) of the effective muon neutrino magnetic moment. A previous analysis of NOvA ND data for a measurement of the effective muon neutrino magnetic moment was presented in a thesis [12], providing a (statistics-only) limit of $\mu_{\nu_\mu} < 15.8 \times 10^{-10} \mu_B$ at 90 % C.L..

Additionally, ν -on- e elastic scattering interactions are used in various other analyses in NOvA, specifically in efforts to constrain the neutrino beam prediction [13] and in the search for Light Dark Matter (LDM) [14]. These analyses developed various tools and methods that can be utilized

in the search for a neutrino magnetic moment.

(TO DO: Possibly include cosmological implications and limits)

In this technical note, we will provide an overview of the theory of neutrino electromagnetic interactions in Sec. 2, focusing on the effective neutrino magnetic moment and its implications for ν -on-e measurements and other theoretical considerations. In Sec. 3, we will discuss the analysis strategy, the signal and background definition, as well as the data and simulation samples and the analysis weights. Following this, in Sec. 4 we will explain the selection of events for this analysis, while in Sec. 5 we will address the relevant systematic uncertainties and in Sec. 6 the statistical methods employed to analyse the results. We will present the results of this analysis in Sec. 7. Finally, section 8 will summarise the findings of this analysis.

2 Theoretical overview

Neutrinos in the SM are massless and electrically neutral particles. However, even SM neutrinos can have electromagnetic interaction through loop diagrams involving charged leptons and the W boson, covered by the neutrino charge radius [1].

In general BSM theories, considering interactions with a single photon as shown on Fig. 1, neutrino electromagnetic interactions can be described by an effective interaction Hamiltonian [15]

$$\mathcal{H}_{em}^{(\nu)}(x) = \sum_{k,j=1}^N \bar{\nu}_k(x) \Lambda_{\mu}^{kj} \nu_j(x) A^{\mu}(x). \quad (1)$$

Here $\nu_k(x)$, $k = 1, \dots, N$, are neutrino fields in the mass basis with N neutrino mass states, Λ_{μ}^{kj} is a general vertex function and $A^{\mu}(x)$ is the electromagnetic field.

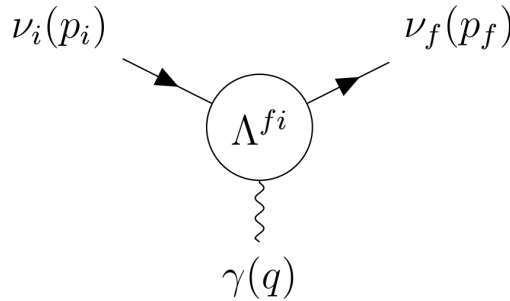


Figure 1: Effective coupling of neutrinos with one photon electromagnetic field.

The vertex function $\Lambda_{\mu}^{fi}(q)$ is generally a matrix and, in the most general case consistent with the SM gauge invariance [16, 17], can be written in terms of linearly independent products of Dirac matrices (γ) and only depends on the four momentum of the photon ($q = p_f - p_i$):

$$\Lambda_{\mu}^{fi}(q) = \mathbb{F}_1^{fi}(q^2) q_{\mu} + \mathbb{F}_2^{fi}(q^2) q_{\mu} \gamma_5 + \mathbb{F}_3^{fi}(q^2) \gamma_{\mu} + \mathbb{F}_4^{fi}(q^2) \gamma_{\mu} \gamma_5 + \mathbb{F}_5^{fi}(q^2) \sigma_{\mu\nu} q^{\nu} + \mathbb{F}_6^{fi}(q^2) \epsilon_{\mu\nu\rho\gamma} q^{\nu} \sigma^{\rho\gamma}, \quad (2)$$

where $\mathbb{F}_i^{fi}(q^2)$ are six Lorentz invariant form factors and δ and ε are the Dirac delta and the Levi-Civita symbols respectively.

Applying conditions of hermiticity ($\mathcal{H}_{em}^{(v)\dagger} = \mathcal{H}_{em}^{(v)}$) and of the gauge invariance of the electromagnetic field, the vertex function can be rewritten as

$$\Lambda_\mu^{fi}(q) = (\gamma_\mu - q_\mu \not{q}/q^2) \left[\mathbb{F}_Q^{fi}(q^2) + \mathbb{F}_A^{fi}(q^2) q^2 \gamma_5 \right] - i\sigma_{\mu\nu} q^\nu \left[\mathbb{F}_M^{fi}(q^2) + i\mathbb{F}_E^{fi}(q^2) \gamma_5 \right], \quad (3)$$

where $\mathbb{F}_Q^{fi}, \mathbb{F}_M^{fi}, \mathbb{F}_E^{fi}$ and $\mathbb{F}_A^{fi}$ are hermitian matrices representing the charge, dipole magnetic, dipole electric and anapole neutrino form factors respectively. It is clear that the vertex function only depends on the square of the four momentum of the photon q^2 . In coupling with a real photon ($q^2 = 0$) these form factors become the neutrino charge and magnetic, electric and anapole moments respectively. Additionally, the neutrino charge radius corresponds to the second term in the expansion of the charge form factor [15].

The above expression can be simplified [18] as

$$\Lambda_\mu^{fi}(q) = \gamma_\mu \left(Q_{\nu_{fi}} + \frac{q^2}{6} \langle r^2 \rangle_{\nu_{fi}} \right) - i\sigma_{\mu\nu} q^\nu \mu_{\nu_{fi}}, \quad (4)$$

where $Q_{\nu_{fi}}$, $\langle r^2 \rangle_{\nu_{fi}}$, and $\mu_{\nu_{fi}}$ are the neutrino charge, effective charge radius (also containing anapole moment), and an effective magnetic moment (also containing electric moment) respectively. This is possible thanks to the similar effects of the neutrino charge radius and the anapole moment, and of the neutrino magnetic and electric moments, on neutrino interactions. Therefore, these are the three neutrino electromagnetic properties (charge, effective charge radius and effective magnetic moment) measured in experiments.

The neutrino electric charge is primarily constrained through measurements of the neutrality of matter and through cosmological observations, which provide much better constraints than neutrino oscillation experiments [15]. On the other hand, the neutrino charge radius would manifest as an increase in the size of the ν -on- e elastic scattering coupling constants, allowing it to be studied in neutrino oscillation experiments such as NOvA. Additionally, the value of the neutrino charge radius in the SM is only an order of magnitude smaller than the current world-leading limits [19] and measuring it could either confirm the validity of neutrino interactions in the SM, or open possibilities to non-standard contributions to neutrino scattering [15]. However, measurement of the neutrino charge radius is not part of this analysis, but may be included in the future re-analysis of the ν -on- e interactions in the NOvA ND.

2.1 Neutrino Electric and Magnetic Dipole Moments

The size and effect of neutrino electromagnetic properties depend on the specific BSM theory applied. Evaluating one loop diagrams in the minimally extended SM with three right-handed Dirac neutrinos gives the first approximation of the electric and magnetic moments, which are now 3×3 matrices with elements:

$$\left. \begin{matrix} \mu_{kj}^D \\ i\varepsilon_{kj}^D \end{matrix} \right\} \simeq \frac{3eG_F}{16\sqrt{2}\pi^2} (m_k \pm m_j) \left(\delta_{kj} - \frac{1}{2} \sum_{l=e,\mu,\tau} U_{lk}^* U_{lj} \frac{m_l^2}{m_W^2} \right), \quad (5)$$

where m_k, m_j are the neutrino masses and m_l are the masses of charged leptons which appear in the loop diagrams [15]. The D superscript denotes Dirac neutrinos and M denotes Majorana neutrinos throughout this section. Also, e is the electron charge, G_F is the Fermi coupling constant, U is the Pontecorvo-Maki-Nakagawa-Sakata (PMNS) neutrino oscillation matrix, and m_W is the mass of the W boson. Higher order electromagnetic corrections were neglected, but can also have a significant contribution, depending on the theory.

It can be seen that Dirac neutrinos have no diagonal electric moments ($\epsilon_{kk}^D = 0$) and their diagonal magnetic moments are approximately

$$\mu_{kk}^D \simeq \frac{3eG_F m_k}{8\sqrt{2}\pi^2} \simeq 3.2 \times 10^{-19} \left(\frac{m_k}{\text{eV}} \right) \mu_B, \quad (6)$$

where μ_B is the Bohr magneton which represents the value of the electron magnetic moment [15]. Neutrino magnetic moments are therefore strongly suppressed by the smallness of neutrino masses, with theoretical predictions in Eq. 6 several orders of magnitude below the reach of current experiments [18].

The transition magnetic moments in the minimally extended SM from Eq. 5 are suppressed with respect to the largest of the diagonal magnetic moments by at least a factor of 10^{-4} due to the m_W^2 in the denominator. The transition electric moments are even smaller due to the mass difference in Eq. 5. Therefore an experimental observation of a magnetic moment larger than in Eq. 6 would indicate physics beyond the minimally extended SM [15, 20].

The suppression of the neutrino magnetic moment by the smallness of its mass can be also expressed in a general case [20]. The ‘natural’ upper limits on the size of the neutrino magnetic moment for any BSM theory that has New Physics (NP) generated at a scale Λ_{NP} can be expressed as [21]

$$\mu_\nu^D (\mu_B) \lesssim 3 \times 10^{-15} \frac{m_\nu^D (\text{eV})}{[\Lambda_{NP} (\text{TeV})]^2}. \quad (7)$$

Therefore for $\Lambda_{NP} \simeq 1 \text{ TeV}$ and $m_\nu^D \lesssim 1 \text{ eV}$ the limit becomes $\mu_\nu^D \lesssim 3 \times 10^{-15} \mu_B$, well below the current experimental capabilities. However, these upper bounds only apply if NP is generated well above the electroweak scale $\Lambda_{EW} \sim 100 \text{ GeV}$ [15].

For Majorana neutrinos, the magnetic and electric form factors (and therefore the magnetic and electric moment matrices) are antisymmetric, thus Majorana neutrinos only have transition moments. The simplest extension of the SM that includes Majorana neutrinos requires either the addition of a Higgs triplet, or right-handed neutrinos together with a Higgs singlet [15]. Neglecting the Feynman diagrams which depend on the model of the scalar sector, the magnetic and electric dipole moments are

$$\left. \begin{matrix} \mu_{kj}^M \\ \epsilon_{kj}^D \end{matrix} \right\} \simeq \mp \frac{3ieG_F}{16\sqrt{2}\pi^2} (m_k \pm m_j) \sum_{l=e,\mu,\tau} \text{Im/Re} [U_{lk}^* U_{lj}] \frac{m_l^2}{m_W^2}, \quad (8)$$

where Im is for μ_{kj}^M and Re is for ϵ_{kj}^D . These are difficult to compare to the Dirac case, due to possible presence of Majorana phases in the PMNS matrices, but it is clear that they have the same

order of magnitude as Dirac transition dipole moments. However, the neglected model dependent contributions can enhance the transition dipole moments for Majorana neutrinos [15].

The natural upper bound on the Majorana magnetic moment is less strict compared to the Dirac neutrinos, due to the antisymmetric nature of Majorana magnetic moment, which requires additional Yukawa couplings in the BSM theory compared to Dirac neutrinos, which can enhance the maximal possible magnetic moment [20]. The limit for Majorana neutrinos can be expressed as

$$\mu_{\alpha\beta}^M (\mu_B) \leq 4 \times 10^{-9} \frac{[m_\nu^M]_{\alpha\beta} (\text{eV})}{[\Lambda_{NP} (\text{TeV})]^2} \frac{m_\tau^2}{|m_\alpha^2 - m_\beta^2|}, \quad \alpha, \beta \in \{e, \mu, \tau\}. \quad (9)$$

Here, the neutrino magnetic moment is expressed in the flavour basis instead of the mass basis, since the charged lepton masses are diagonal here. The two bases are related by

$$\mu_{ij} = \sum_{\alpha\beta} \mu_{\alpha\beta} U_{\alpha i}^* U_{\beta j}. \quad (10)$$

and the effect of the neutrino magnetic moment on neutrino interactions does not depend on the choice of the basis[22].

These considerations imply, that if a magnetic moment $\mu \gtrsim 10^{-15} \mu_B$ were measured, neutrinos are almost certainly Majorana particles [20].

2.1.1 Effective neutrino magnetic moment

As mentioned above, the neutrino magnetic moment measured in experiments is the so-called effective neutrino magnetic moment, which is a combination of electric and magnetic dipole moments and depends on the neutrino source and oscillations. In the ultra-relativistic limit, the effective neutrino magnetic moment is

$$\mu_{\nu_l}^2 (L, E_\nu) = \sum_j \left| \sum_k U_{lk}^* e^{\mp i \Delta m_{kj}^2 L / 2E_\nu} (\mu_{jk} - i \epsilon_{jk}) \right|^2, \quad (11)$$

where the minus sign in the exponent is for neutrinos and the plus sign for antineutrinos [15]. Therefore, the only difference between the effective neutrino and antineutrino magnetic moment is in the phase induced by neutrino oscillations. For experiments with baselines short enough that neutrino oscillations would not have time to develop ($\Delta m^2 L / 2E_\nu \ll \sim 1$), such as the NOvA ND, the effective magnetic moment is the same for neutrinos and antineutrinos and is independent of the neutrino energy.

Since the effective magnetic moment depends on the initial neutrino flavour, it is different for experiments studying neutrinos from different sources. Additionally, experiments such as solar neutrino experiments, need to include matter effects on the neutrino oscillations. Therefore the reports on the value (or upper limit) of the effective neutrino magnetic moment are not directly comparable between different types of neutrino experiments.

2.2 Measuring the Neutrino Magnetic Moment

The most sensitive method to measure the neutrino magnetic moment is the low energy elastic scattering of (anti)neutrinos on electrons [15]. The schematic diagram for this interaction is shown in Fig. 2, where the recoil electron's kinetic energy is defined as $(T_e = E_{e'} - m_e)$ and the recoil angle with respect to the incoming neutrino beam (θ) is shown.

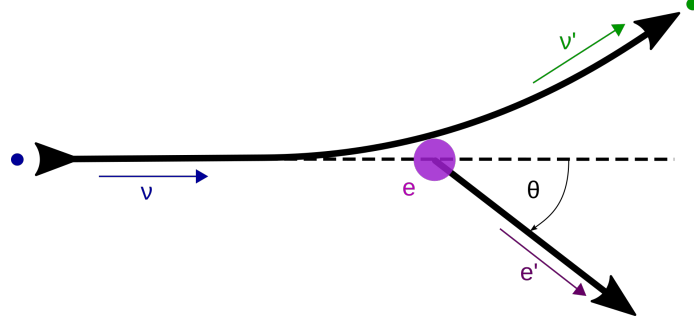


Figure 2: Neutrino-on-electron elastic scattering diagram

Since the ν -on- e interaction is governed by simple $2 \rightarrow 2$ kinematics, it can be shown that

$$(P_V - P_{e'})^2 = (P_{V'} - P_e)^2, \quad (12)$$

$$m_V^2 + m_e^2 - 2E_V E_{e'} + 2E_V p_{e'} \cos \theta = m_{V'}^2 + m_e^2 - 2E_{V'} m_e. \quad (13)$$

From the energy conservation

$$E_V + m_e = E_{V'} + E_{e'} = E_{V'} + T_e + m_e \Rightarrow E_{V'} = E_V - T_e \quad (14)$$

it follows that

$$E_V p_{e'} \cos \theta = E_V E_{e'} - E_{V'} m_e = E_V (T_e + m_e) - (E_V - T_e) m_e = T_e (E_V + m_e), \quad (15)$$

$$\cos \theta = \frac{E_V + m_e}{E_V} \sqrt{\frac{T_e^2}{E_{e'}^2 - m_e^2}} = \frac{E_V + m_e}{E_V} \sqrt{\frac{T_e^2}{T_e^2 + 2T_e m_e}}. \quad (16)$$

And finally

$$\cos \theta = \frac{E_V + m_e}{E_V} \sqrt{\frac{T_e}{T_e + 2m_e}}. \quad (17)$$

Which can be rearranged to get

$$T_e = \frac{2m_e E_V^2 \cos^2 \theta}{(E_V + m_e)^2 - E_V^2 \cos^2 \theta}. \quad (18)$$

The electron's kinetic energy is therefore constrained as

$$T_e \leq \frac{2E_V^2}{2E_V + m_e}, \quad (19)$$

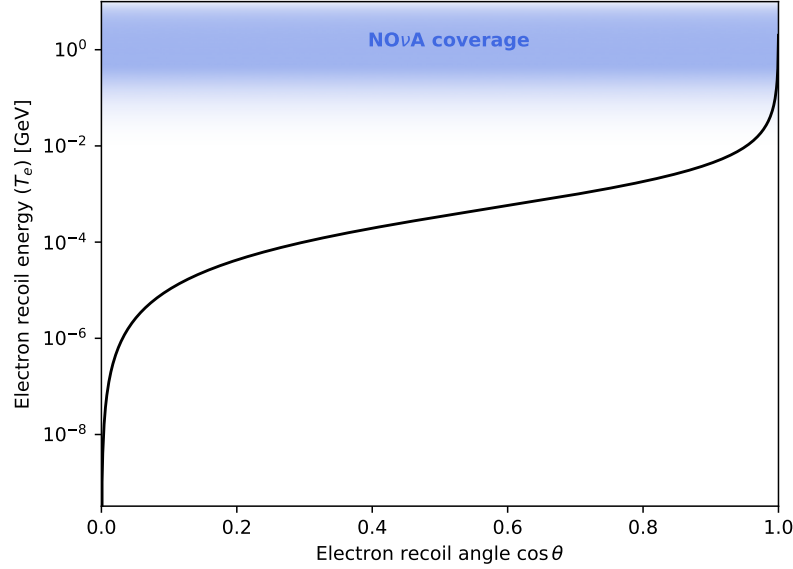


Figure 3: Relation between the recoil electron's kinetic energy and angle for the ν -on-e elastic scattering. The coverage of the NOvA detectors for measuring the electron recoil energy is shown in blue. Only very forward electrons are therefore recorded in NOvA.

160 which corresponds to the limit $\cos \theta \rightarrow 1$ when the recoil electron goes exactly forward in the
 161 incident neutrino direction, as depicted in Fig. 3.

Considering $E_\nu \sim \text{GeV}$, it is useful to approximate $\frac{m_e^2}{E_\nu^2} \rightarrow 0$. Additionally, considering only very small electron recoil angles, meaning $\theta^2 \cong (1 - \cos^2 \theta)$, applied to Eq. 17 results in

$$T_e \theta^2 \cong T_e \left(1 - \left(\frac{E_\nu + m_e}{E_\nu} \right)^2 \frac{T_e}{T_e + 2m_e} \right) = T_e \left(1 - \left(1 + \frac{2m_e}{E_\nu} \right) \frac{T_e}{T_e + 2m_e} \right), \quad (20)$$

therefore

$$T_e \theta^2 \cong \frac{2m_e T_e}{T_e + 2m_e} \left(1 - \frac{T_e}{E_\nu} \right) = 2m_e \left(\frac{1}{1 + \frac{2m_e}{T_e}} \right) \left(1 - \frac{T_e}{E_\nu} \right), \quad (21)$$

and finally

$$T_e \theta^2 \cong 2m_e \left(1 - \frac{T_e}{E_\nu} \right) < 2m_e. \quad (22)$$

162 This is a strong limit that very clearly distinguishes the ν -on-e elastic scattering events from
 163 other similar interactions involving single electron (mainly the ν_e Charged Current (CC) interac-
 164 tions).

2.2.1 Neutrino Magnetic Moment Cross Section

In the ultra-relativistic limit, the neutrino magnetic moment interaction flips the neutrino helicity, while the SM weak interaction conserves it, which means it is possible to add the two contributions to the total ν -on- e cross section incoherently (without interference terms) [15]:

$$\frac{d\sigma_{\nu-on-e}}{dT_e} = \left(\frac{d\sigma_{\nu-on-e}}{dT_e} \right)_{\text{SM}} + \left(\frac{d\sigma_{\nu-on-e}}{dT_e} \right)_{\text{MAG}}. \quad (23)$$

The SM contribution can be expressed as [23, 15]:

$$\left(\frac{d\sigma_{\nu-on-e}}{dT_e} \right)_{\text{SM}} = \frac{2G_F^2 m_e}{\pi} \left\{ g_1^2 + g_2^2 \left(1 - \frac{T_e}{E_\nu} \right)^2 - g_1 g_2 \frac{m_e T_e}{E_\nu^2} \right\}, \quad (24)$$

where the coupling constants g_1 and g_2 differ between neutrino flavours and between neutrinos and antineutrinos. Their values are:

$$g_1^{\nu_e} = g_2^{\bar{\nu}_e} = \sin^2 \theta_W + 1/2, \quad g_2^{\nu_e} = g_1^{\bar{\nu}_e} = \sin^2 \theta_W, \quad (25)$$

$$g_1^{\nu_{\mu,\tau}} = g_2^{\bar{\nu}_{\mu,\tau}} = \sin^2 \theta_W - 1/2, \quad g_2^{\nu_{\mu,\tau}} = g_1^{\bar{\nu}_{\mu,\tau}} = \sin^2 \theta_W, \quad (26)$$

where $\sin^2 \theta_W \cong 0.23$.

The total SM cross section, and therefore the number of SM ν -on- e interactions, depends on the neutrino energy and the minimum measured electron recoil energy. However, in general the cross section for ν_e is about 2.5 times larger than for the $\bar{\nu}_e$, about 6 times larger than for $\nu_{\mu/\tau}$ and about 7 times larger than for $\bar{\nu}_{\mu/\tau}$.

The neutrino magnetic moment contribution is [24, 15]:

$$\left(\frac{d\sigma_{\nu-on-e}}{dT_e} \right)_{\text{MAG}} = \frac{\pi \alpha^2}{m_e^2} \left(\frac{1}{T_e} - \frac{1}{E_\nu} \right) \left(\frac{\mu_{\nu_l}}{\mu_B} \right)^2, \quad (27)$$

where α is the fine structure constant and μ_{ν_l} is the effective magnetic moment of ν_l . The total cross section now only depends on the neutrino energy and on the effective magnetic moment, but is the same for neutrinos and antineutrinos.

The comparison of the SM and the neutrino magnetic moment differential cross sections is shown in Fig.4. Whereas the SM cross section is approximately uniform for $T_e \rightarrow 0$, the neutrino magnetic moment cross section rises to infinity. However, this reach is limited by the experimental capabilities of detecting electrons with very low energies. The (possible) NOvA coverage is shown with a shaded blue region, with current capability reaching $T_e = 0.5 \text{ GeV}$. Future analyses might extend this reach to lower T_e , with the lowest possible detectable electron recoil energy $T_{e,min} \approx 0.01 \text{ GeV}$.

Calculating the ratio of the neutrino magnetic moment and the SM cross sections, as shown in Fig. 5, can serve as a proxy to estimate the number of neutrino magnetic moment events in relation to the predicted number of SM events, if the E_ν and T_e are known. Additionally, comparing the ratio of the total cross sections can reveal the expected total number of neutrino magnetic

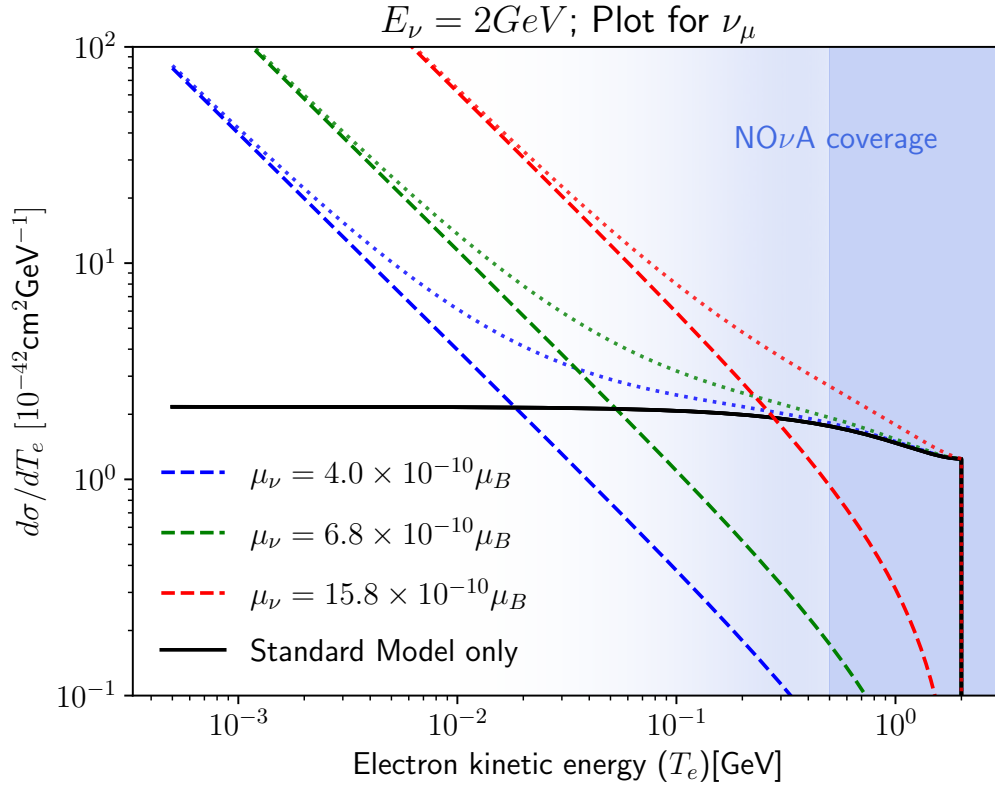


Figure 4: Comparison of the neutrino magnetic moment (coloured) and the SM (black) cross sections for the ν -on- e elastic scattering. Different colours depict different values of the neutrino magnetic moment, with red corresponding to the previous NOvA measurement, green the LSND result, and blue a possible ultimate NOvA sensitivity, as discussed in the introduction to this chapter. Dashed lines are the individual cross sections and dotted lines are the added total cross section with the standard model contribution. NOvA coverage of electron recoil energies is shown in shaded blue.

moment events as a function of the predicted number of SM events. Considering $E_\nu = 2 \text{ GeV}$, $\mu_\nu = 6.8 \times 10^{-10} \mu_B$ (current best limit for ν_μ from LSND), and integrating differential cross sections for ν_μ in Eq. 24 and 27 from $T_{e,min}$ to $T_{e,max} \rightarrow 2 \text{ GeV}$ results in

$$\frac{\sigma_{\text{MAG}}}{\sigma_{\text{SM}}} \approx \begin{cases} 0.035 & T_{e,min} = 0.5 \text{ GeV}, \\ 0.14 & T_{e,min} = 0.01 \text{ GeV}. \end{cases} \quad (28)$$

Therefore, at the current NOvA detection capabilities, there are about 0.035 times as many neutrino magnetic moment ν -on- e events than SM ones. This can be compared with the expected statistical uncertainty on the SM background, which in the case of Poisson distributed events is the square root of the number of predicted events. Consequently, it is possible to assess the minimal number of SM ν -on- e events necessary for the magnetic moment signal to be detected above the SM background (without considering systematic uncertainties) as

$$N_{\text{SM}} > 1/0.035^2 \approx 816. \quad (29)$$

181 However, this approximation is calculated only for one value of E_ν , but can be used to assess the
182 sensitivity of the experiment.

As can be seen in Fig. 4 and Fig. 5, the magnetic moment contribution exceeds the SM contribution for low enough T_e . This can be approximated as [15]:

$$T_e \lesssim \frac{\pi^2 \alpha^2}{G_F^2 m_e^3} \left(\frac{\mu_\nu}{\mu_B} \right)^2 \simeq 2.9 \times 10^{19} \left(\frac{\mu_\nu}{\mu_B} \right)^2 [\text{MeV}], \quad (30)$$

183 which does not depend on the neutrino energy. Therefore, experiments sensitive to lower energetic
184 electrons are significantly more sensitive to the neutrino magnetic moment.

185 3 Analysis overview

186 Our analysis strategy for measuring the effective muon neutrino magnetic moment in the NOvA
187 ND is based on comparing the total number of reconstructed and selected events in data with the
188 prediction. The predicted events consist of the signal, which depends on the size of the effective
189 muon neutrino magnetic moment, and of the background, which corresponds to the SM-only (null)
190 hypothesis without any neutrino magnetic moment. We define the signal as true ν -on- e elastic
191 scattering interactions, created with the use of the neutrino magnetic moment cross section instead
192 of the SM cross section, as described in Sec. 2.2. Additionally, the signal events are required
193 to have their true interaction vertex contained within the ND to exclude events originating from
194 outside of the detector.

195 The ν -on- e interactions are also used in the ND group's analysis to constraint the neutrino
196 beam prediction [13], which relies on the precise theoretical knowledge of the ν -on- e interaction
197 cross section. They compare the total number of recorded ν -on- e interactions, with background
198 subtracted based on a data/Monte Carlo (MC) comparison in a sideband region, to the prediction.
199 Since the number of ν -on- e events should only depend on the normalization of the neutrino beam,

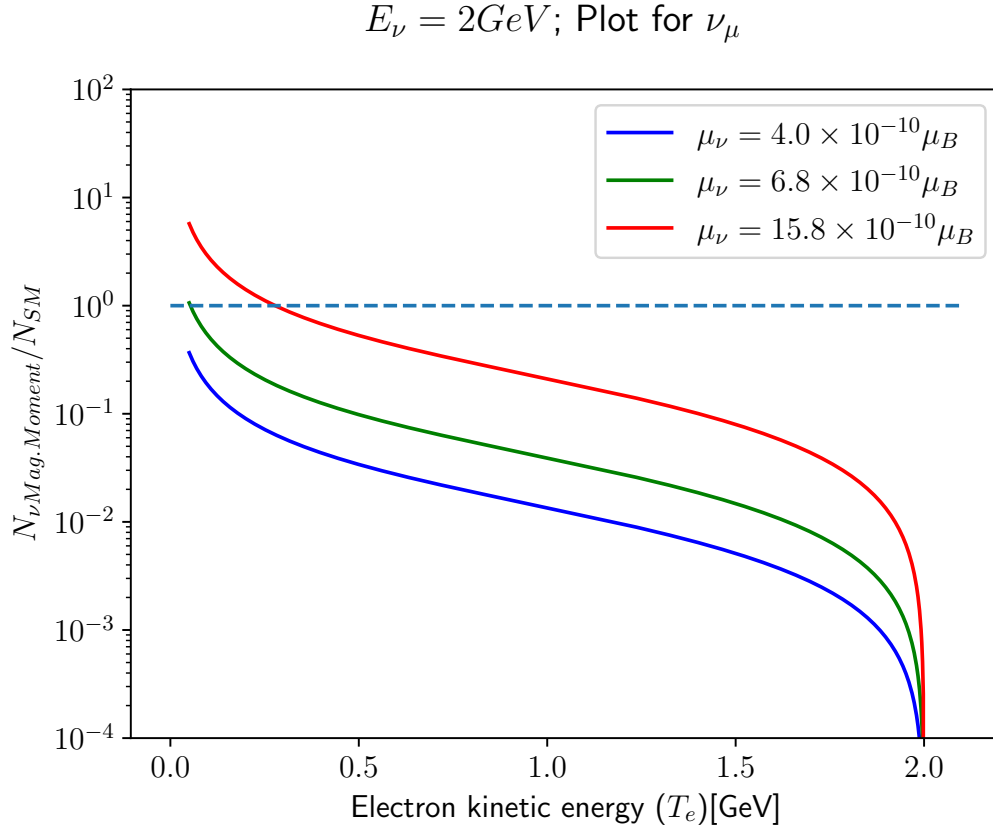


Figure 5: Ratio of the neutrino magnetic moment cross section to the SM cross section for the ν -on- e elastic scattering of 2GeV ν_μ . Different colours depict different effective muon neutrino magnetic moment values, with red corresponding to the previous NOvA measurement, green the LSND result, and blue a possible ultimate NOvA sensitivity, as discussed in the introduction to this chapter.

this analysis should give us a precise validation, or correction, of the neutrino flux normalization. There has been a large amount of work going into this analysis, including making special samples, weights, event classifiers, developing a dedicated event selection, or developing a background subtraction method, among others. To save time and analysis effort, we have taken most of these work at face value and applied it to the neutrino magnetic moment analysis. This will help us get the first good estimate of NOvA’s capabilities to constraint (or measure) the neutrino magnetic moment.

The same detector signature of a single forward going electron shower, as present in the ν -on-e events, is also present in the LDM analysis [14]. This analysis is using a similar event selection to select the LDM events as the ND group, only without the final $E\theta^2$ cut (see Sec. 4). However, instead of simply comparing the total event counts, the LDM analysis is using a CAFAna-based fitting framework to fit for the possible LDM signal in a distribution of electron recoil energy multiplied by electron recoil angle squared.

The data used in this analysis were collected from the start of NOvA ND data taking on the August 22nd, 2014, until February 3rd, 2021. This is the ND data that were used in the latest NOvA neutrino oscillations result [25], with an additional year. Although more data have been collected since February 2021, they are still being processed and are not available at the time of writing this thesis. The total exposure of the data sample is approximately 13.8×10^{20} Protons On Target (POT). This exposure is used throughout this chapter to scale the predicted distributions and number of events.

For this analysis we are using near detector CAF samples with the standard Production 5.1 reconstruction. The SAMWEB definition for the data sample is:

```
prod_caf_R20-11-25-prod5.1reco.l_nd_numi_fhc_full_v1_goodruns.
```

We are following the standard data blinding procedure and have not looked at any data events until the analysis passes the full collaboration review.

We are using the standard NOvA Prod5.1 ND simulation, with Geant4 v4.10.4.p02 [26] used for the neutrino flux prediction and GENIE v3.0.6 [27] with tune N1810j_02_11a for the neutrino interactions. This is the GENIE version that experienced the ‘*Genie Skew*’ (TO DO: *Describe the genie skew fix properly and reference, maybe docdb:53691 or something newer*). The list of CAFAna simulatio samples is in Tab. 1.

The simulation was created with approximately four times larger statistics than the data to limit statistical uncertainties from simulation with total exposure for the simulation being approximately 55.4×10^{20} POT. For the systematic uncertainty studies only a portion of this full sample is used, specifically 19.3×10^{20} POT.

Analysis Weights

Corrections for known limitations in the simulation are applied in the form of analysis weights applied to each event based on how it is affected by specific variations in the simulation. This includes the corrections for the neutrino beam prediction based on the external measurements used by the Package to Predict the Flux (PPFX), and, for the non- ν -on-e background only, also

Table 1: SAMWEB definitions of the nominal Near Detector Prod5.1 simulation samples.

Nominal:

prod_caf_R20-11-25-prod5.1reco.a_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_v1

Systematically shifted samples:

prod_caf_R20-11-25-prod5.1reco.e_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_calibup_v1_batch2

prod_caf_R20-11-25-prod5.1reco.e_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_calibdown_v1_batch2

prod_caf_R20-11-25-prod5.1reco.f_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_ckvup_v1_batch2

prod_caf_R20-11-25-prod5.1reco.f_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_ckvdown_v1_batch2

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_lightlevelup_v1_batch2

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_lightlevelup_v1_batch2

prod_caf_R20-11-25-prod5.1reco.f_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_detectorageing_v1_batch2

prod_caf_R20-11-25-prod5.1reco.f_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_calibshape_v1_batch2

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08
_full_mcnp_v1_batch2

239 the internal and external measurements that constrain the neutrino interaction prediction inside
240 GENIE.

241 **Radiative Correction Weight**

242 The cross section corrections are not applied to the ν -on-e events, as they are assumed to be known
243 precisely from theory. However, the GENIE MC simulation only considers tree-level SM ν -on-e
244 interactions [13], as described in Sec. 2.2, and doesn't account for any higher order terms, which
245 are described by radiative corrections.

Radiative corrections can be expressed by two adjustments to the tree-level SM ν -on-e cross section [28]. First, the values of the weak coupling constants are changed as [29]

$$g_1^{V_e} \rightarrow 0.7276, \quad g_1^{V_\mu} \rightarrow -0.2730, \quad g_2 \rightarrow 0.2334. \quad (31)$$

Second, there are additional terms added to the cross section equation. Considering only one-loop corrections, the full ν -on- e cross section can be expressed as¹ [30]

$$\left(\frac{d\sigma_{\nu-on-e}}{dy}\right)_{\text{Rad. Corr.}} = \frac{G_F^2 s}{\pi} \left\{ g_1^2 \left(1 + \frac{\alpha}{\pi} X_1\right) + g_2^2 (1-y)^2 \left(1 + \frac{\alpha}{\pi} X_2\right) - g_1 g_2 \frac{m_e y}{E_\nu} \right\}, \quad (32)$$

where

$$y = \frac{T_e + E_\gamma}{E_\nu}, \quad (33)$$

$s = 2E_\nu m_e + m_e^2$ is the Mandelstam variable,

$$X_1 = -\frac{2}{3} \log\left(\frac{2yE_\nu}{m_e}\right) + \frac{y^2}{24} - \frac{5y}{12} - \frac{\pi^2}{6} + \frac{23}{72} \quad (34)$$

and

$$X_2 = -\frac{2}{3} (1-y)^2 \log\left(\frac{2yE_\nu}{m_e}\right) - \frac{y^2}{18} - \frac{\pi^2}{6} (1-y)^2 - \frac{2y}{9} + \frac{23}{72}. \quad (35)$$

In practice, we are using the implementation of radiative corrections as a weight, as created by Yiwen Xiao [13], where each true ν -on- e event is weighted by a ratio

$$\text{weight}_{\text{Rad. Corr.}}(E_\nu, T_e) = \left(\frac{d\sigma_{\nu-on-e}}{dy}\right)_{\text{Rad. Corr.}} \bigg/ \left(\frac{d\sigma_{\nu-on-e}}{dy}\right)_{\text{GENIE 3}}; \quad (36)$$

246 Neutrino Magnetic Moment Signal as a Weight

Analogically to the radiative correction weight, it is possible to create a neutrino magnetic moment weight as a ratio between the neutrino magnetic moment and the SM differential cross sections for the ν -on- e interactions. This can then serve to predict the number of ν -on- e events created by the neutrino magnetic moment interaction (which make up the signal), without the need for an additional simulation. This is possible thanks to the theoretically very well understood properties of the ν -on- e interaction, as described in Sec. 2.2. Therefore, the signal sample is created from the true ν -on- e sample, with the magnetic moment weight applied. The weight has a form:

$$\text{weight}_{\nu\text{Mag. Mom.}}(E_\nu, T_e) = \left(\frac{d\sigma_{\nu-on-e}}{dy}\right)_{\nu\text{Mag. Mom.}} \bigg/ \left(\frac{d\sigma_{\nu-on-e}}{dy}\right)_{\text{GENIE 3}}, \quad (37)$$

where

$$\left(\frac{d\sigma_{\nu-on-e}}{dy}\right)_{\nu\text{Mag. Mom.}} = E_\nu \left(\frac{d\sigma_{\nu-on-e}}{dT_e}\right)_{\nu\text{Mag. Mom.}}. \quad (38)$$

¹There is technically a third correction term X_3 by the $g_1 g_2$ term, which is however negligible for $E_\nu \sim \text{GeV}$.

Enhanced ν -on-e sample

Due to the relatively low cross section of the ν -on-e interaction, the nominal simulation sample contains very few ν -on-e events, which could result in a significant statistical uncertainty from simulation. To avoid this, we are using the ν -on-e-enhanced simulation sample created by Wenjie [13], which is mainly made up of ν -on-e events with a total exposure of 1.72×10^{24} POT. There are a few non- ν -on-e background events overlaid on top of the ν -on-e events to properly account for the possible reconstruction effects of the pileup of neutrino interactions in a single spill [13], since in the real detector, the hits from the true ν -on-e interaction can be clustered together into another interaction, or additional hits can be clustered together into the ν -on-e event. To save up on unnecessary disk space and processing usage, the enhanced ν -on-e sample does not include any cross section related parameters and variables, as the ν -on-e interaction is assumed to be known exactly from theory. Therefore, we do not apply cross section weights or account for cross section systematic uncertainties for ν -on-e events.

(**TO DO: Describe the technical details of the nuone sample**) We are using the overlaid and filematched samples for consistency. The CAF files have incorrect POT, which is $3.6995434\text{e}+20$. This needs to be corrected to their correct POT of $1.7209423\text{e}+24$ (this is for filematched sample). We only have systematic samples for the absolute calibration, cherenkov and light level. This is because the effect of ageing, calibration shape, and neutrons should be negligible for ν -on-e events.

The list of the nu-on-e sample definitions is in table 2.

Table 2: SAMWEB definitions for the enhanced ν -on-e samples.

Nominal:

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_v1_nuone_overlay

Systematically shifted samples:

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_calibup_v1_nuone_overlay

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_calibdown_v1_nuone_overlay

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_ckvup_v1_nuone_overlay

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_ckvdown_v1_nuone_overlay

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_lightlevelup_v1_nuone_overlay

prod_caf_R20-11-25-prod5.1reco.g_nd_genie_N1810j0211a_nonswap_fhc_nova_v08_full_lightleveldown_v1_nuone_overlay

Enhanced ν_e CC MEC sample

The cross section tuning procedure in NOvA applies large weights to Meson Exchange Current (MEC) events in some parts of the parameter space. However, after the full event selection (Sec. 4) only a small number of MEC events remain in the detector. This was shown to be an issue especially for the ν_e CC MEC events [13]. Applying large tuning corrections to a small number of events results in large statistical fluctuations. To avoid this, we are using another special sample created by Yiwen Xiao [13] with an enhanced number of ν_e CC MEC events, following the same procedure as for the ν -on-e-enhanced sample, with an exposure of 1.99×10^{24} POT.

(**TO DO: Describe the technical details**) Incorrect POT is 4.7334120e+23 and correct POT is 1.9880340e+24. This is for filematched samples. There is a limitation of the sample in the q3-q0 parameter space. Need to list it or just reference the technote.

The list of the ν_e CCMEC enhanced sample definitions is in table 3.

Table 3: SAMWEB definitions of the ν_e CC MEC background sample.

Nominal:

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_filematch

Systematically shifted samples:

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_CalibUp_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_CalibDown_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_CkvUp_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_CkvDown_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_LLP_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_LLP_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_Aging_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_CalibShape_Combined_240124_filematch

YiwenXiao_NueCCMEC_Single_NJobs7500_CAF_NonSwap_MCNP_Combined_240124_filematch

A summary of the simulation samples and analysis weights for the four different types of signal and background components is shown in Tab. 4. In the following chapter, the ν_e CC MEC background is added into the ‘Other background’ sample, even though it is created from a separate simulation.

4 Event Selection

We are trying to select low energy neutrino-on-electron events, which are characterised by a single very forward going electron shower. Since these are the same events as are used in the Near

Table 4: Overview of the simulation samples and analysis weight used for the different signal and background components.

Signal type	Sample	Weight
Signal	Enhanced ν -on-e	Flux & ν Mag. Moment
ν -on-e background	Enhanced ν -on-e	Flux & Rad. Corr.
ν_e CC MEC background	Enhanced ν_e CC MEC	Flux & Cross Sec.
Other background	Nominal ND	Flux & Cross Sec.

Detector group’s analysis to constraint the neutrino beam prediction with a neutrino-on-electron events [13], we have taken their event selection without changes. This is to save time and analysis efforts, but also to get a first good estimation of NOvA’s capabilities to constraint (or measure) neutrino magnetic moment. Almost the same selection is also used in the Light Dark Matter analysis [14], only without the final $E\theta^2$ cut.

We explain the motivation behind each cut of the event selection and discuss their effect on the neutrino magnetic moment events below. We also consider possible improvements to the event selection for a future (re-)analysis.

The code to make plots and analyse the event selection is located in the NuMagMomentAna/NuMMAAnalysis/EventSelection directory. The signal definition is in NuoneCuts.h and the cut values in the EventCVNCuts.h/cxx scripts from the NuMagMomentAna/Cuts directory.

Signal Definition

To define the signal and background samples we use the true information listed in Tab. 5. The neutrino magnetic moment signal definition is the same as that for the neutrino-on-electron background with the neutrino magnetic moment weight applied (explained in Sec. ??).

The signal definitions use the kMode variable instead of the kIntType, which is now deprecated (ref.: various Slack conversations). Mode 10005 denotes only the ν -on-e events and is only available for the enhanced ν -on-e samples, while mode 5 denotes all electron scattering events, including inverse muon decay interactions. That is why we had to add a requirement of an electron in the final state. Mode 10 denotes all Meson Exchange Current (MEC) events.

We are using the ND group’s [13] signal definition including a requirement for the true vertex to lie within their fiducial volume (defined below). This is the cut name NDNuoneFiducial. This is in contrast with the LDM analysis [14], which uses the ana:kVtxIsContained cut instead, which has a looser boundary. This choice has only a negligible effect on the final number of selected events and only affects the selection efficiency.

Spill Quality Selection

To ensure good data quality, we apply the following criteria on the run, subrun and spill level. (TO DO: Add a description of the spill quality cuts (only applied to data))

Table 5: Overview of signal and background definitions.

Signal	<code>kMode== 10005 && NDNuoneFiducial</code>
ν -on-e background	<code>kMode== 10005 && NDNuoneFiducial</code>
ν_e CC MEC background	<code>!(kMode== 5 && kElInFinState && NDNuoneFiducial) && (kIsCC && kIsNue && kMode == 10)</code>
Other background	<code>!(kMode== 5 && kElInFinState && NDNuoneFiducial) && !(kIsCC && kIsNue && kMode == 10)</code>

Basic Reconstruction Quality Selection

(**TO DO: Split the pre-selection into basic reco quality and basic event selection**) Pre-selection cuts include basic quality cuts that remove events with invalid vertex reconstruction and events with no reconstructed prongs, as shown on Fig. 6. In pre-selection we also include a cut on the time difference between the mean times of the "current" slice and of the slice closest in time, which should be > 25 ns. This ensures that ... (**TO DO: This is to remove slicing failures that split the two gamma from a π^0 into two slices that resemble electron signal. Should we remove them now though?**).

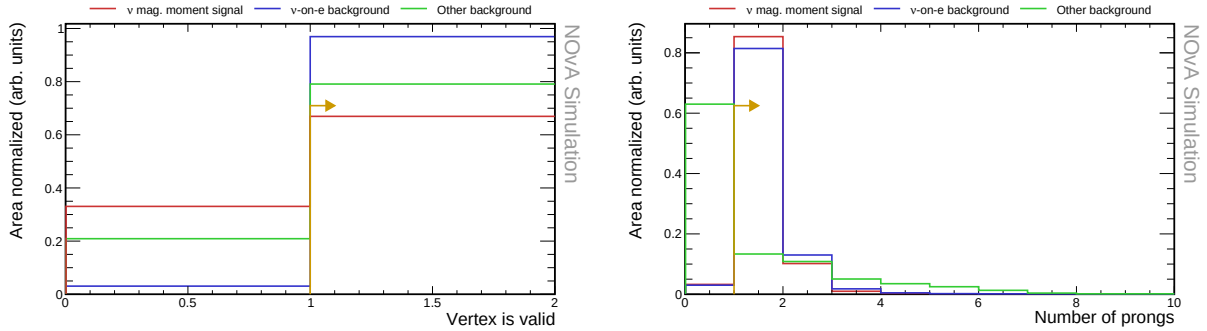


Figure 6: Relative comparison of signal, ν -on-e background, and other background events for basic pre-selection variables. No cuts were applied to make these plots. Gold lines show the cut values for the shown variables.

Basic Event Selection

The pre-selection cuts have been kept from the ν_e CC analysis with loosened cut values (**TO DO: find a reference for this analysis**). They also remove the obvious ν_μ CC interactions by requiring that the length of the longest prong is < 800 cm, number of planes crossed by the longest prong is < 120 , and the summed number of cells for all prongs in the slice is < 600 . Relative comparison of signal, ν -on-e background, and other background distributions for the pre-selection variables is shown on Fig. 7.

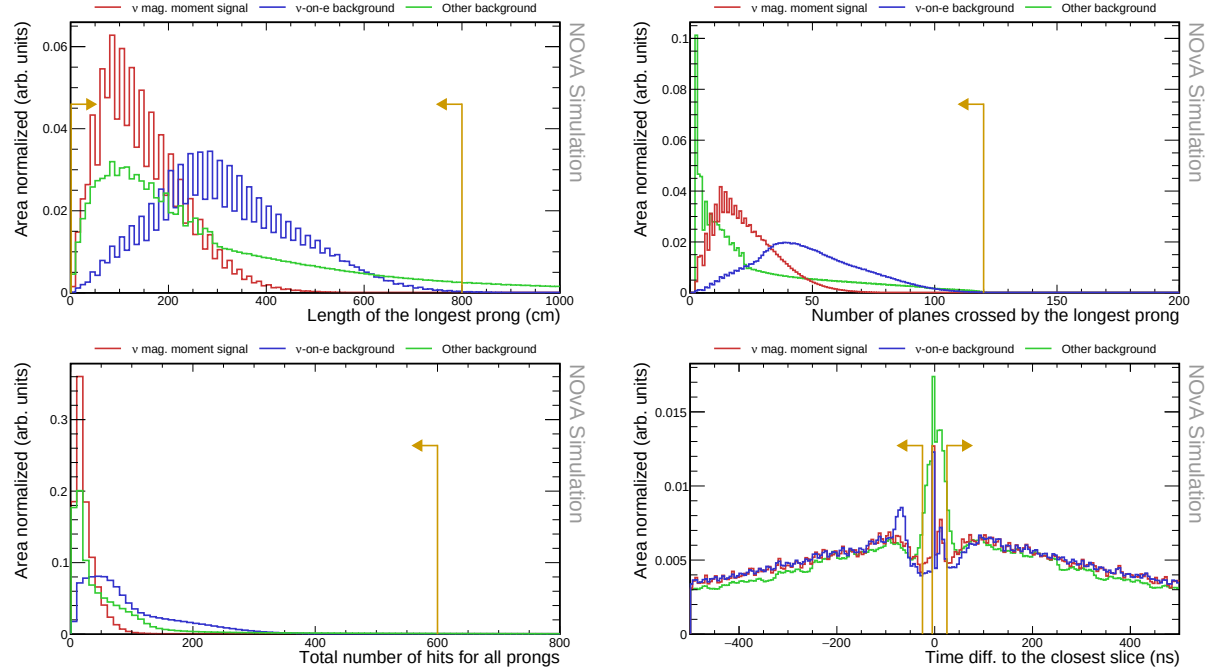


Figure 7: Relative comparison of signal, v-on-e background, and other background events for pre-selection variables. Cuts on VtxIsValid and number of prongs were applied to make these plots. Gold lines show the cut values for the shown variables.

Fiducial and containment cuts

(TO DO: Describe what does the fiducial cut do) We require that the reconstructed vertex is contained within the following volume: $-185 < Vtx_X < 175$, $-175 < Vtx_Y < 175$, $95 < Vtx_Z < 1095$ cm.

To ensure all the energy is contained within the detector and to remove events originating outside of the detector (rock muons), we require that the extreme positions of hits for all prongs in the slice are within the following volume: $-190 < \min_X, \max_X < 180$, $-180 < \min_Y, \max_Y < 190$, $105 < \min_Z, \max_Z < 1275$ cm.

Single particle requirement

To selection events with a single particle we require that the fraction of energy contained in the most energetic shower is > 0.8 , that the summed energy of all cells (above threshold and within ± 8 planes from the vertex) outside of the most energetic shower is < 0.02 GeV, and that the distance between the vertex and the start of the primary shower is < 20 cm.

Shower energy cut

(TO DO: discuss the energy cut, should this be removed? What is the effect on the event count? Why was this included in the first place (the identifiers are not as strong for lower energies - is

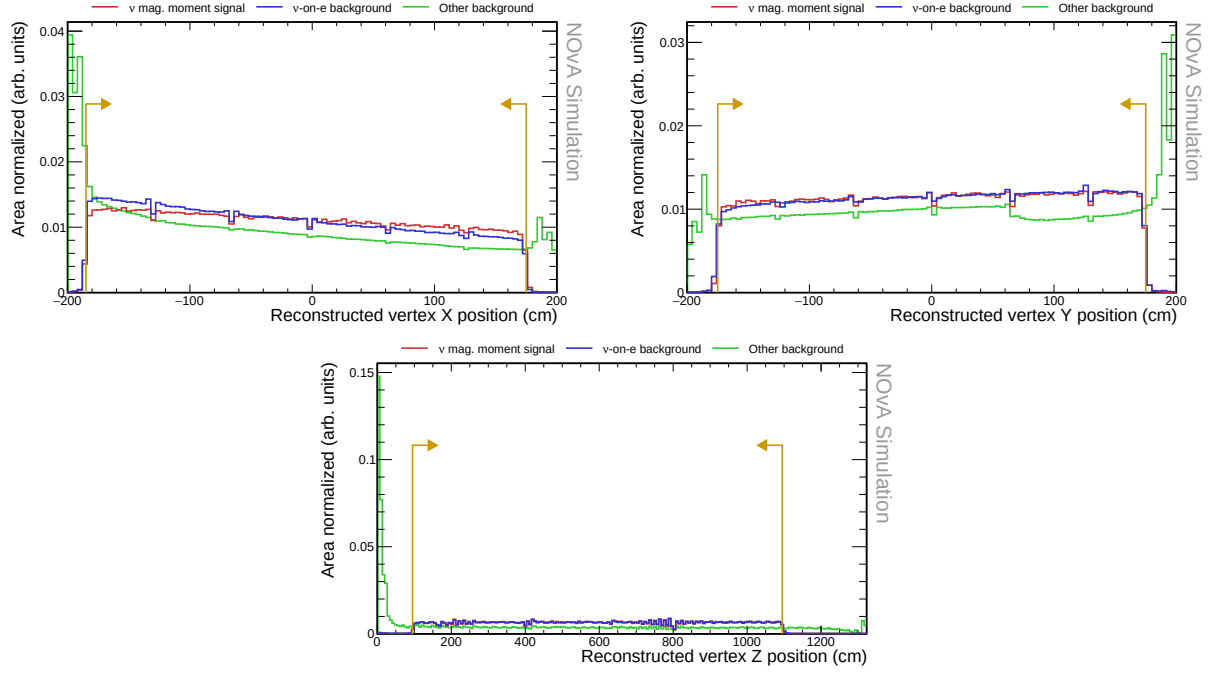


Figure 8: Relative comparison of signal, ν -on- e background, and other background events for the reconstructed vertex. No cuts were applied to make these plots. Gold lines show the cut values that create the fiducial volume.

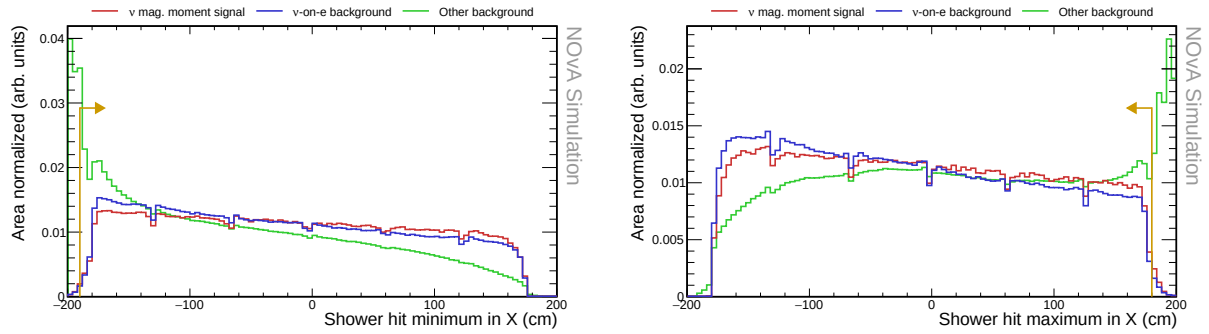


Figure 9: Relative comparison of signal, ν -on- e background, and other background events for the minimum and maximum position of the reconstructed shower along the X axis. Pre-selection and fiducial cuts were applied to make these plots. Gold lines show the values of the containment cuts.

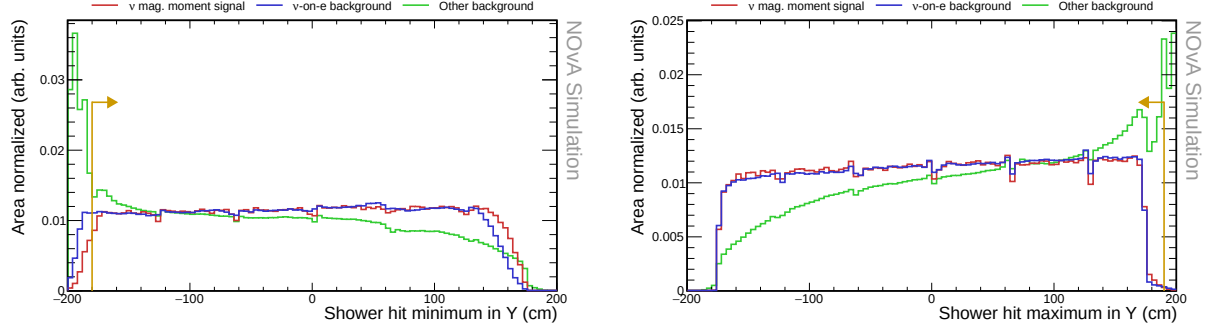


Figure 10: Relative comparison of signal, ν -on-e background, and other background events for the minimum and maximum position of the reconstructed shower along the Y axis. Pre-selection and fiducial cuts were applied to make these plots. Gold lines show the values of the containment cuts.

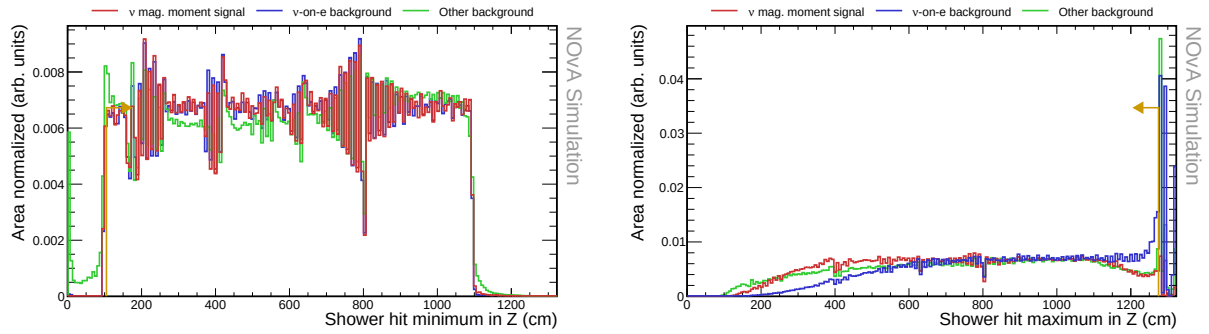


Figure 11: Relative comparison of signal, ν -on-e background, and other background events for the minimum and maximum position of the reconstructed shower along the Z axis. Pre-selection and fiducial cuts were applied to make these plots. Gold lines show the values of the containment cuts.

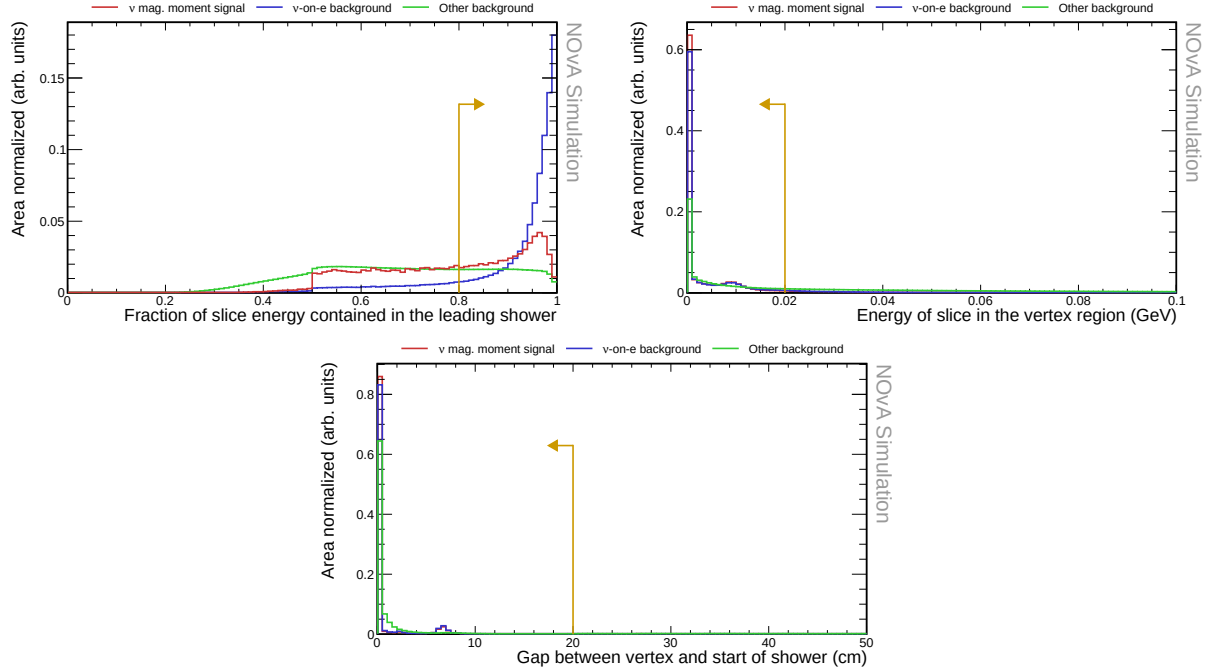


Figure 12: Relative comparison of signal, ν -on-e background, and other background events for the reconstructed vertex. No cuts were applied to make these plots. Gold lines show the cut values that create the fiducial volume.

this true though? - also there are further unexplored backgrounds that would need to be further studied and explore. Maybe depends on where would we move the cut...)) The calorimetric energy of the primary shower is required to be within $0.5 < E_{cal} < 5$ GeV.

Event classifiers

We are using two event classifiers based on convolution neural network that were developed specifically to identify ν -on-e interactions. The first one (NuoneID) is trained to select ν -on-e events and the second one (Epi0ID) is trained on the events passing the NuoneID to reject the π^0 background. Our selection requires that NuoneID > 0.73 and that Epi0ID > 0.92 .

(TO DO: reference theory for the kinematics of nuone scattering) We require that the product of reconstructed energy of the primary shower and the square of its angle from the Z axis is $E_{cal}\theta^2 < 0.005 \text{ GeV} \times \text{rad}^2$.

(TO DO: Add plots of distributions of the event selection variables with two columns. LHS shows no cuts applied and RHS shows all previous cuts applied)

Using the many plots below that show the effect of each of the cuts on the signal and all background events. (For signal we are showing NuMM=...)

(TO DO: Describe the cutflow tables below) The final event count and efficiency of each of the cuts is shown on the table 6. Table 14 shows the dissemination of background into the individual components.

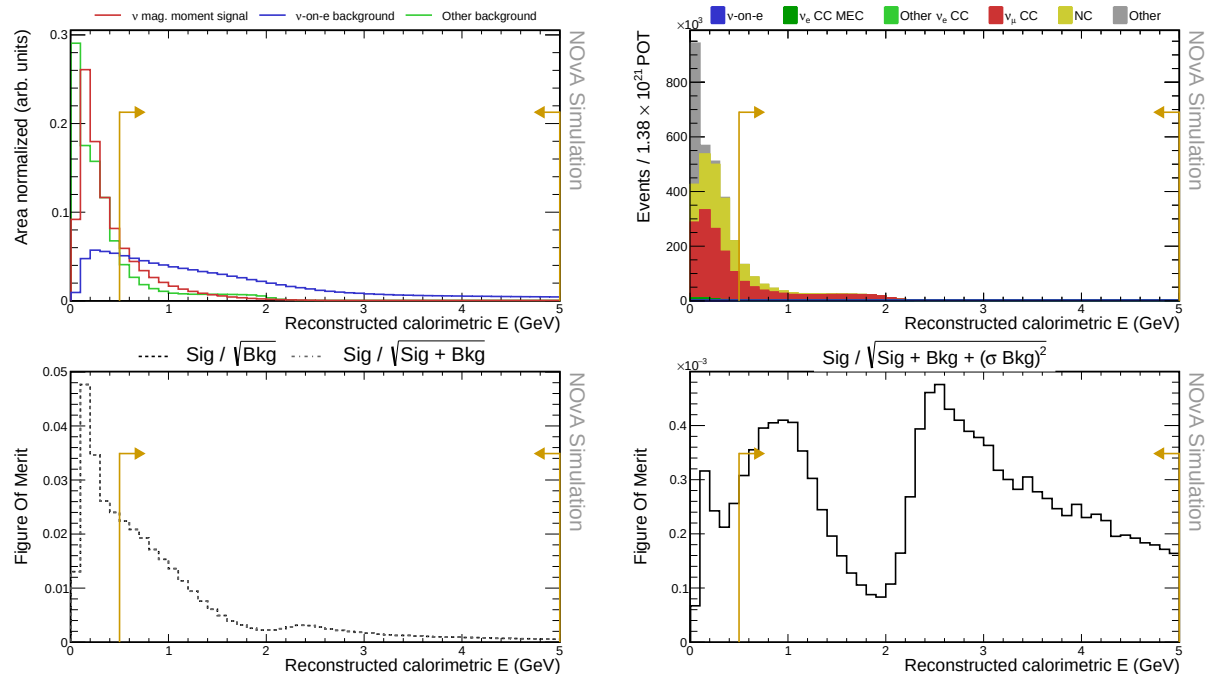


Figure 13: Relative comparison of signal, ν -on- e background, and other background events for the reconstructed vertex. No cuts were applied to make these plots. Gold lines show the cut values that create the fiducial volume.

(TO DO: Add a discussion of possible improvements on the event selection on its limitations - mostly for the analysis review committee) From here we can see that ... Maybe what can be improved is... This can likely be improved upon by specifically selection low energy events and removing the cut on the reconstructed shower energy.

5 Systematic uncertainties

(TO DO: Describe the main systematic uncertainties. Add plots showing their effect on the NuMM events. Possibly with different event selection variables as X axes. Also show the final table with the percentages summed) Plots showing combined uncertainties for signal and backgrounds. Maybe also some interpolations. Table of systematic uncertainties on the event count.

5.1 Normalization systematics

(TO DO: Describe the normalization systematics (or just remove this if not using them in the end) Should we include normalization systematics? Would that make any difference? There's a POT scaling uncertainty which is very small (find out exactly how small).

In the fitting experiment normalization uncertainties would probably not make any difference whatsoever, but in the counting experiment they might be important?

5.2 Neutrino flux systematics

(TO DO: Describe the flux uncertainties. Describe the PCA. Describe the difference between what the ND group is doing and what we're doing) Using the PCA vs using the PPF universes+beam transport separately. Plots of energy showing shifts for signal and backgrounds separately

(TO DO: understand differences with ND and 3F methods)

This is mainly a normalization. Discuss how to use the fact that ν -on-e events can be used (and are used) to constraint the beam uncertainty. Would the counting experiment still be valid then? Maybe if we made another sideband sample...

5.3 Detector systematics

(TO DO: Make plots of energy showing shifts for signal and backgrounds separately) Reference for the Prod5.1 detector systematics is docdb 53225

5.4 Cross section systematics

(TO DO: Describe the XSec systs) Only for the non nu-on-e background. Assuming the nu-on-e events (including the signal events) are precisely known.

Table 6: Event selection cutflow table

Selection	ν Mag. Moment signal			ν -on-e background			Other background		
	N_{sig}	ϵ^{N-1}	ϵ (%)	N_{IBkg}	ϵ^{N-1}	ϵ (%)	N_{Bkg}	ϵ^{N-1}	ϵ (%)
No Cut	269.77	100	100	3.43×10^3	100	100	2.96×10^8	100	100
Vtx Is Valid	180.58	66.94	66.94	3.33×10^3	96.94	96.94	2.34×10^8	79.09	79.09
N Prongs	174.69	96.74	64.76	3.23×10^3	96.99	94.02	8.66×10^7	37.00	29.27
Png Length	174.67	99.99	64.75	3.22×10^3	99.64	93.68	7.67×10^7	88.56	25.92
N Planes	174.67	100	64.75	3.22×10^3	99.98	93.67	7.67×10^7	99.98	25.92
N Cells	174.67	100	64.75	3.22×10^3	99.98	93.65	7.42×10^7	96.78	25.08
Closest Slc	169.82	97.22	62.95	3.14×10^3	97.54	91.35	6.95×10^7	93.68	23.49
Fiducial	167.72	98.76	62.17	3.09×10^3	98.41	89.89	3.59×10^7	51.71	12.15
Cont.	159.37	95.02	59.08	2.48×10^3	80.43	72.30	1.38×10^7	38.35	4.66
ShwE Frac.	150.37	94.35	55.74	2.42×10^3	97.59	70.56	8.82×10^6	63.97	2.98
Vtx E	142.29	94.63	52.74	2.18×10^3	90.16	63.62	4.15×10^6	47.07	1.40
Shw Gap	137.96	96.96	51.14	2.09×10^3	95.58	60.80	3.25×10^6	78.34	1.10
Shw E	37.13	26.92	13.76	1.36×10^3	65.10	39.58	6.25×10^5	19.21	0.21
Nuoneid	29.48	79.39	10.93	940.21	69.18	27.38	2.42×10^4	3.88	8.19×10^{-3}
Epi0id	22.51	76.35	8.34	749.93	79.76	21.84	1.47×10^4	60.75	4.97×10^{-3}
$E\theta^2$	19.74	87.73	7.32	675.02	90.01	19.66	84.15	0.57	2.84×10^{-5}
$E\theta^2$ (sb)	2.74	-	1.01	74.30	-	2.16	1.01×10^3	-	3.43×10^{-4}
No ShwE	37.62	-	13.94	782.67	-	22.79	238.79	-	8.07E-05

Plots of energy showing shifts for signal and backgrounds separately

5.5 Fitting framework

(TO DO: Describe the fitting framework, ideally with an example plot, maybe with some arrows)

How does the fitting framework work? It's based on the framework developed by Mu Wei for the Light Dark Matter analysis (ref.) which was developed together (is this fair?). Basic description of the framework.

Also this framework is used for both LDM and NuMM together. It is trivial to simply switch between including the NuMM or LDM in it. This was done to save space in creating predictions since our backgrounds are exactly the same (or at least they should be...). Theoretically this could be separated into two different frameworks.

- `<NDPredictionSingleElectron>` Prediction class which holds the LDM as a special 2-D spectrum (not used for NuMM), and NuMM, ν -on-e background, ν_e CC MEC background and other background as simple 1-D spectra. Also scaling each spectra by...
- `<NDPredictionSystSingleElectron>` class derived from `PredictionInterp` that takes in the `NDPredictionSingleElectron` and applies systematic shifts to it. Includes the interpolation/extrapolation between the systematic shifts.
- `FitVariables` and what do they do
- Fitter which does exactly what... What are the parameters of the fit? What are the results/outputs?

6 Statistical Analysis

6.1 Resolution and binning

(TO DO: Add the energy resolution and binning plots) The electron energy and angle distributions and resolutions. Are we going to fit in E, Th, or ETh2? Is there something else?

Show plots of Reco V True for both energy and angle. (Should I show it with or without the energy cut?). Also show the resolution plots.

6.2 Fitting framework

(TO DO: Describe the fitting framework, ideally with an example plot, maybe with some arrows)

How does the fitting framework work? It's based on the framework developed by Mu Wei for the Light Dark Matter analysis (ref.) which was developed together (is this fair?). Basic description of the framework.

Also this framework is used for both LDM and NuMM together. It is trivial to simply switch between including the NuMM or LDM in it. This was done to save space in creating predictions

since our backgrounds are exactly the same (or at least they should be...). Theoretically this could be separated into two different frameworks.

- `<NDPredictionSingleElectron>` Prediction class which holds the LDM as a special 2-D spectrum (not used for NuMM), and NuMM, ν -on-e background, ν_e CC MEC background and other background as simple 1-D spectra. Also scaling each spectra by...
- `<NDPredictionSystSingleElectron>` class derived from `PredictionInterp` that takes in the `NDPredictionSingleElectron` and applies systematic shifts to it. Includes the interpolation/extrapolation between the systematic shifts.
- `FitVariables` and what do they do
- Fitter which does exactly what... What are the parameters of the fit? What are the results/outputs?

7 Results / Fake data studies

[to do]: talk about the shape-only versus normalisation-only strategies for the differential versus single-bin analyses.

I think it might be a good idea to talk about this already in the analysis overview section.

I need to also discuss fake data studies. Should I talk about this now? Probably

7.1 Counting experiment

Single bin analysis - total numbers of signal and backgrounds predicted. Possible background corrections and its effect on the background uncertainties.

7.2 Binned experiment

Plots showing delta chi2 for stats and systs separately. Plot showing the chi2 with limits. Maybe talk about using different binning and variables and the effect.

Also need to somehow quantify the various fitter settings, like different seeding values, different range of profiling. Should I talk about Feldman-Cousins?

7.3 Template fit

(TO DO: Describe the fitting framework, ideally with an example plot, maybe with some arrows)
How does the fitting framework work? It's based on the framework developed by Mu Wei for the Light Dark Matter analysis (ref.) which was developed together (is this fair?). Basic description of the framework.

Also this framework is used for both LDM and NuMM together. It is trivial to simply switch between including the NuMM or LDM in it. This was done to save space in creating predictions

since our backgrounds are exactly the same (or at least they should be...). Theoretically this could be separated into two difference frameworks.

- `<NDPredictionSingleElectron>` Prediction class which holds the LDM as a special 2-D spectrum (not used for NuMM), and NuMM, ν -on-e background, ν_e CC MEC background and other background as simple 1-D spectra. Also scaling each spectra by...
- `<NDPredictionSystSingleElectron>` class derived from `PredictionInterp` that takes in the `NDPredictionSingleElectron` and applies systematic shifts to it. Includes the interpolation/extrapolation between the systematic shifts.
- `FitVariables` and what do they do
- Fitter which does exactly what... What are the parameters of the fit? What are the results/outputs?

7.3.1 Sensitivities and limits

8 Conclusion

(TO DO: Report the limit with its uncertainty)

(TO DO: Very briefly discuss differences with current world limit and how the techniques differ)

(TO DO: Very briefly summarise expectations for future measurements.)

Acronyms

ν -on-e neutrino-on-electron (interaction). 1–3, 6–17

BSM Beyond Standard Model. 1–5

C.L. Confidence Level. 1

CC Charged Current. 7, 16, 17

LDM Light Dark Matter. 1, 12

MC Monte Carlo. 10, 13

MEC Meson Exchange Current. 16, 17

ND Near Detector. 1, 3, 5, 10, 12, 17

NOvA NuMI Off-axis ν_e Appearance (experiment). 1, 3, 5, 7–12, 16

NP New Physics. 4, 5

PMNS Pontecorvo-Maki-Nakagawa-Sakata. 4

POT Protons On Target. 12, 15, 16

PPFX Package to Predict the Flux. 12

SM Standard Model. 1–4, 8–11, 13, 14

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Figure 14: Event selection cutflow table for background components

Selection	$V_e \text{CC MEC}$		$V_e \text{CC Other}$		$V_{\mu} \text{CC}$		NC		Other	
	N	ϵ^{N-1}	N	ϵ^{N-1}	N	ϵ^{N-1}	N	ϵ^{N-1}	N	ϵ^{N-1}
No Cut	3.50×10^4	100	3.23×10^6	100	2.24×10^8	100	3.40×10^7	100	3.49×10^7	100
Vtx Is Valid	3.27×10^4	93.58	2.62×10^6	81.14	1.99×10^8	89.02	2.57×10^7	75.55	6.53×10^6	18.70
N Prongs	2.74×10^4	83.76	1.39×10^6	53.05	6.75×10^7	33.89	1.51×10^7	58.57	2.65×10^6	40.51
Png Length	2.73×10^4	99.79	1.37×10^6	98.53	5.77×10^7	85.56	1.49×10^7	99.07	2.64×10^6	99.87
N Planes	2.73×10^4	99.99	1.37×10^6	99.99	5.77×10^7	99.98	1.49×10^7	100	2.64×10^6	100
N Cells	2.73×10^4	99.99	1.28×10^6	93.49	5.59×10^7	96.82	1.44×10^7	96.34	2.64×10^6	100
Closest Slc	2.73×10^4	99.79	1.21×10^6	94.25	5.33×10^7	95.40	1.35×10^7	94.17	1.43×10^6	54.22
Fiducial	1.39×10^4	51.12	6.30×10^5	52.10	2.60×10^7	48.77	8.25×10^6	60.99	1.05×10^6	73.53
Cont.	9.32×10^3	66.82	2.63×10^5	41.72	7.64×10^6	29.38	4.96×10^6	60.15	9.12×10^5	86.62
ShwE Frac.	9.20×10^3	98.70	1.95×10^5	74.39	4.82×10^6	63.10	2.97×10^6	59.78	8.28×10^5	90.81
Vtx E	5.92×10^3	64.33	6.05×10^4	30.96	1.97×10^6	40.79	1.36×10^6	45.75	7.62×10^5	92.03
Shw Gap	5.50×10^3	92.91	4.62×10^4	76.40	1.58×10^6	80.18	1.06×10^6	77.78	5.69×10^5	92.03
Shw E	3.62×10^3	65.81	1.12×10^4	24.15	4.38×10^5	27.80	1.71×10^5	16.15	1.28×10^3	0.23
Nuoneid	1.40×10^3	38.63	2.11×10^3	18.89	1.17×10^4	2.66	8.99×10^3	5.27	66.43	5.17
Epl0id	1.14×10^3	81.78	1.61×10^3	76.40	7.17×10^3	61.52	4.76×10^3	52.94	29.47	44.36
$E\theta^2$	15.13	1.32	39.00	2.42	1.21×10^{-3}	3.85×10^{-6}	20.91	0.44	6.15×10^{-5}	1.43×10^{-6}
$E\theta^2$ (sb)	386.16	-	306.55	-	9.48×10^{-3}	7.40×10^{-5}	149.93	-	4.41×10^{-4}	1.79×10^{-5}
No ShwE	15.54	-	69.61	-	2.15×10^{-3}	3.06×10^{-5}	75.67	-	2.22×10^{-4}	2.72E-05